

01. Foundation

19 September 2026 | Ready for review

A working local interface and API, with repository execution intentionally disabled. This is the first approval checkpoint, not a completed repair agent.

What you can review now

- Next.js interface with real setup status and clear empty states.
- Spring Boot API protected by a local server-to-server token.
- Docker diagnostics that report the actual unavailable daemon.
- Local startup, shutdown, configuration, and safety checks.
- A documented plan for repository overview and change-explanation reports.

Project location

/home/tejas-ng/Desktop/coding/projects/code base Doctor

Approval rule

Each milestone is implemented, debugged, and documented in a PDF before asking for your confirmation. Milestone 2 has not started. All later product features remain disabled.

This report excludes secret values. Its file list covers project source, configuration, scripts, documentation, and generated local configuration. Dependency caches and generated build output are summarized as directories, not thousands of individual vendor files.

Project folder structure

Snapshot of the delivered foundation. Names are relative to the project folder. Existing .git, .agents and .codex metadata was not modified.

```
code base Doctor/
|-- .dockerignore
|-- .env
|-- .env.example
|-- .gitignore
|-- PROJECT_CONTEXT.md
|-- README.md
|-- backend/
|   |-- pom.xml
|   |-- src/
|       |-- main/
|           |-- java/
|               |-- dev/
|                   |-- codebasedoctor/
|                       |-- DoctorApplication.java
|                       |-- api/
|                           |-- FoundationController.java
|                           |-- config/
|                               |-- LocalApiGuard.java
|                               |-- model/
|                                   |-- Models.java
|                                   |-- sandbox/
|                                       |-- DockerSandbox.java
|                   |-- resources/
|                       |-- application.properties
|               |-- test/
|                   |-- java/
|                       |-- dev/
|                           |-- codebasedoctor/
|                               |-- FoundationApiTest.java
|-- docker-compose.yml
|-- docker/
|   |-- backend.Dockerfile
|   |-- frontend.Dockerfile
|-- docs/
|   |-- DOCUMENTATION_FEATURE.md
|   |-- MILESTONES.md
|   |-- MILESTONE_1_VALIDATION.md
|-- frontend/
|   |-- .env.local
|   |-- AGENTS.md
|   |-- CLAUDE.md
|   |-- app/
|       |-- api/
|           |-- [...path]/
|               |-- route.ts
|       |-- globals.css
|       |-- layout.tsx
|       |-- page.tsx
|   |-- components/
|       |-- workbench.tsx
|-- lib/
```

Folder structure, continued

```
| | |-- types.ts
| | |-- next-env.d.ts
| | |-- next.config.ts
| | |-- package-lock.json
| | |-- package.json
| | |-- public/
| | |-- `-- favicon.svg
| |-- `-- tsconfig.json
|-- milestone-reports/
| |-- Milestone-01-Foundation.pdf
| |-- README.md
|-- `-- scripts/
| |-- dev.py
| |-- dev.sh
| |-- setup-local.sh
| |-- smoke.py
| |-- test.sh
| |-- `-- test_dev.py
```

Generated directories: `.runtime/m2` (Maven dependencies), `frontend/node_modules` (npm packages), `frontend/.next` (compiled UI), `backend/target` (Java build and test output), `scripts/__pycache__` (Python bytecode). They are ignored by Git.

Created files and their purpose

All entries below belong to Milestone 1. Planned-only surfaces are identified explicitly. No generated report or repair feature is presented as operational.

.dockerignore

Excludes secrets, dependencies, build output, and local runtime data from Docker build contexts.

.env

Private local application settings and generated API token; values intentionally excluded from this report.

.env.example

Documents environment variable names and example configuration without real credentials.

.gitignore

Keeps credentials, installed dependencies, build artifacts, caches, and runtime data out of Git.

PROJECT_CONTEXT.md

Records project scope, current milestone, architecture decisions, verified progress, and next steps.

README.md

Explains local setup, startup, foundation capabilities, checks, and the milestone approval workflow.

backend/pom.xml

Defines the Java 21 Spring Boot application, dependency versions, test dependencies, and Maven packaging.

backend/src/main/java/dev/codebasedoctor/DoctorApplication.java

Starts the Spring Boot API and discovers application components.

backend/src/main/java/dev/codebasedoctor/api/FoundationController.java

Provides authenticated health and empty job-list endpoints; rejects repository execution during Milestone 1.

backend/src/main/java/dev/codebasedoctor/config/LocalApiGuard.java

Requires a strong configured API token, authenticates API calls, and rejects oversized requests.

backend/src/main/java/dev/codebasedoctor/model/Models.java

Defines shared job, analysis, finding, plan, change, event, and verification data structures for later workflows.

backend/src/main/java/dev/codebasedoctor/sandbox/DockerSandbox.java

Reads bounded Docker CLI/daemon diagnostics while keeping repository execution explicitly disabled.

backend/src/main/resources/application.properties

Binds the API to loopback, sets request/error-handling limits, and maps environment-based configuration.

backend/src/test/java/dev/codebasedoctor/FoundationApiTest.java

Checks authentication, configuration validation, request limits, truthful health state, and disabled job creation.

docker-compose.yml

Defines optional local UI/API containers with restricted privileges and resources; mounts no Docker socket.

docker/backend.Dockerfile

Builds the backend JAR and runs it as a non-root user in a Java runtime image.

docker/frontend.Dockerfile

Builds the frontend with all source directories and runs its standalone production server as a non-root user.

docs/DOCUMENTATION_FEATURE.md

Specifies future repository overview and change reports, evidence rules, Markdown downloads, and acceptance checks.

docs/MILESTONES.md

Lists nine delivery milestones with debugging requirements and explicit user approval before each advance.

docs/MILESTONE_1_VALIDATION.md

Records exact Milestone 1 checks, debugging fixes, observed results, and unverified limitations.

frontend/.env.local

Private frontend-server token and local backend address; credentials are not browser-public and values are omitted.

frontend/AGENTS.md

Next.js-generated development guidance for future coding agents; created automatically by the dev server.

frontend/CLAUDE.md

Next.js-generated reference to the local agent guidance file.

frontend/app/api/[...path]/route.ts

Proxies allowlisted API routes to the loopback backend using the server-only token; validates origin, host, body type, and size.

frontend/app/globals.css

Provides the workbench visual theme, responsive layout, and component styling.

frontend/app/layout.tsx

Defines the page document, application metadata, and global stylesheet import.

frontend/app/page.tsx

Renders the main Codebase Doctor workbench page.

frontend/components/workbench.tsx

Provides the interactive foundation workbench and renders real local application health and unavailable workflow states.

frontend/lib/types.ts

Defines frontend types for health, jobs, analysis, findings, changes, plans, events, and verification results.

frontend/next-env.d.ts

Provides generated Next.js type declarations for TypeScript.

frontend/next.config.ts

Enables standalone packaging, sets browser security headers, and uses the TypeScript compiler API with type checking enabled.

frontend/package-lock.json

Locks resolved frontend dependency versions for repeatable installation.

frontend/package.json

Defines Next.js/React/TypeScript dependencies and development, build, start, and type-check commands.

frontend/public/favicon.svg

Provides the application-specific browser icon.

frontend/tsconfig.json

Defines TypeScript compiler checks and frontend import aliases.

milestone-reports/Milestone-01-Foundation.pdf

User-facing Milestone 1 report containing the project folder structure, each file purpose, validation results, and current limitations.

milestone-reports/README.md

Explains the milestone PDF archive and the required structure, file-purpose inventory, and debugging evidence for every milestone.

scripts/dev.py

Starts the trusted local UI/API, parses configuration as data, and shuts down child process groups.

scripts/dev.sh

Prepares local configuration and launches the Python development-process supervisor.

scripts/setup-local.sh

Creates private local environment files with a generated API token and refuses conflicting existing frontend configuration.

scripts/smoke.py

Exercises local HTTP health, disabled job creation, origin/host protections, route handling, and request-size limits.

scripts/test.sh

Runs the configuration-parser test, backend tests, and frontend production build.

scripts/test_dev.py

Checks that configuration stays literal and cannot override host command lookup, and that service process groups are terminated.

Debugging and verification

Backend tests - PASS

Five Spring Boot tests passed with zero failures or errors. They cover authentication, disabled jobs, accurate health state, weak token rejection, and request-size limits.

Local setup and shutdown tests - PASS

Two Python regression tests passed: `.env` is literal allowlisted data, and service process groups are terminated.

Frontend build and TypeScript - PASS

The production build compiled, completed type checking, prerendered its static routes, and produced standalone output. A separate TypeScript check also exited successfully.

Live UI-to-API checks - PASS

Eight real HTTP smoke checks passed: health, empty jobs, disabled job creation, foreign Origin, foreign Host, cross-site fetch, unknown route, and oversized body.

Browser verification - PASS

The browser showed API Connected and the actual Docker unavailable message. Workspace, Documentation, Activity, and mode controls worked. Analyze stayed disabled. No browser console errors were captured. Mobile and wider layouts were inspected.

Secret and configuration checks - PASS

The actual local API token was absent from production browser bundles. Both local environment files have mode 0600 permissions. Shell script syntax checks passed.

Docker Compose configuration - PASS; STARTUP NOT RUN

Compose configuration validation succeeded. Container startup and rootless/cgroup enforcement could not be tested because the Docker daemon is inaccessible in this session.

Problems fixed during this milestone

- Replaced executing `.env` as shell code with literal parsing of approved keys.
- Replaced parent-process-only shutdown with process-group cleanup and a regression test.
- Aligned the Docker frontend recipe with standalone output and the component source directory.
- Corrected a default component import caught by the TypeScript check.
- Switched to the supported TypeScript compiler API because this environment returned empty CLI capture output; retained full type checking.

Safety boundary and remaining work

- No external repository was ingested, built, modified, committed, or pushed during this milestone.
- Repository execution is blocked in the UI and at the backend, even for an authenticated caller.
- The browser does not receive the local API token. Requests from foreign browser origins and unapproved hosts are rejected.
- Configuration is parsed as literal data. Stopping the launcher terminates its service process groups.
- Docker Compose configuration was validated, but container startup was not tested because this session cannot access the Docker daemon.
- No Groq or GitHub key was provided. This milestone makes no model or publishing requests.
- Docker-based execution is future work. Containers share a kernel; the later sandbox will require explicit verification and must not be described as an absolute PC safety guarantee.

Documentation feature planned for the product

A later approved milestone will generate a repository overview, main-change summary, per-file reasons, architecture grounded in source, the actual diff, before/after build and test results, and unresolved concerns. It will mark unrun checks and AI suggestions clearly. This product feature is separate from the milestone PDFs supplied during development.

Next confirmation

Review this foundation and its checks. With your confirmation, Milestone 2 will implement bounded public GitHub repository intake and safe job workspaces. This report does not authorize the next milestone or any repository execution.